Independent reconstruction of the coastal adaptation model of Wong et al. (2025): continuing research report

Research-agent record prepared for Sophia Ludtke

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Abstract

Coastal cost distributions depend on both sea-level response and the joint uncertainty assigned to climate, exposure, and adaptation. This project examines whether the results of Wong et al. (2025) can be reconstructed from scholarly documentation without consulting the authors' implementation or computational outputs. We recover component equations, implement numerical kernels in Julia, and compare manufactured cases with analytical limits and an independent SciPy integration. Continued source recovery supplies coupled climate and Antarctic dynamics, a closed carbon-cycle alternative, and independent observational inputs. A temperature-only calibration passes convergence diagnostics but fails held-out temperature and upper-ocean heat checks. These failures motivate a separate temperature–heat calibration; they are retained as evidence of model limitations. The calculations verify heat conservation, ice-response convergence, flood-loss integration, and economic accounting. They also expose a finite-step residual in the printed emissions recurrence and a lower-bound discrepancy in an Antarctic temperature transform. These are findings about written equations, not evidence of defects in the inaccessible original software. A complete independent joint calibration and the target coastal input bundle were not recovered. Consequently, none of the thirteen numerical target figures is independently reproduced. An original workflow diagram, verified component calculations, explicitly labeled alternative experiments, a complete figure register, and recovery requirements document ongoing work. This report does not conclude that all scientifically feasible reproduction work is exhausted.

1 Objective and evidentiary boundary

Wong et al. [1] estimate global coastal adaptation costs and losses conditional on warming in 2100. Their model connects emissions, carbon cycling, temperature and ocean heat, global and local sea level, and segment-level coastal decisions. The scientific objective here is to reproduce the resulting conditional distributions from the documented mechanisms. It is not sufficient to draw curves resembling the published figures or to reproduce marginal parameter ranges while discarding their dependence.

The independence restriction permits scientific papers, their supplementary methods, and exogenous inputs from independent providers. Published figures and printed summary statistics are comparison evidence only. No source code, processed input bundle, posterior sample, simulation file, or plotting script from the target authors was opened or used. An initial broad literature search displayed incidental repository-index snippets, including a data-loading excerpt. Those results were discarded, and subsequent searches focused on scientific documents and provider records. This exposure is retained in the research log rather than represented as complete isolation from search results.

The recovered sources include the target article and its ten-figure supplement, the SNEASY climate description [3], its carbon-cycle reference [4], BRICK and MimiBRICK descriptions [5, 6], the DAIS paper [7], and the CIAM mathematical supplement [8]. The December 2024 revision of the cited emissions preprint [2] was recovered through public scholarly file-version metadata. Its current DOI landing page serves a later 2025 version. Since the target citation does not pin a revision, this historical manuscript remains a version-qualified specification. Stable acquisition URLs and hashes are recorded in `docs/SOURCES.csv`.

2 Original experiment and recovered interfaces

The target source ensemble contains 100,000 states of the world, with 5,000 used for the coastal analysis. Global warming in 2100, relative to 1850–1900, defines four bins: at most 1.5 °C, above 1.5 and at most 2.5 °C, above 2.5 and at most 3.5 °C, and above 3.5 °C. Global sea level in 2100 uses a 1995–2014 reference. The baseline socioeconomic scenario is SSP4; SSP2 provides a sensitivity case. Monetary outcomes are in 2010 US dollars with a 4% annual discount rate [1].

The coastal model uses 12,148 DIVA segments and decadal calculations from 2010 through 2150. Planning intervals are 2010–2050, 2060–2100, and 2110–2150. Protection and retreat standards are associated with return periods of 1, 10,100, 1,000, and 10,000 years, alongside no adaptation. The paper specifies nine adaptation combinations. The complete 2022 scholarly supplement resolves the menu: protection at 10–10,000 years and retreat at 1–10,000 years. The nominal one-year level is annual maximum high tide, obtained for the GTSR experiment by scaling the DINAS one-year level by the GTSR/DINAS 100-year ratio. It is not the GEV quantile at probability zero.

The 2022 main methods specify first-period cost minimization, persistence of the selected strategy and return level, and later height augmentation. Actual adaptation already in place is not globally observed; the original calculation initializes an optimal adaptation state. Reconstructing this spin-up requires the same exposure and extreme-water inputs as the forward model. It cannot be inferred from a global cost total.

Figure1 identifies the implemented kernels and the missing links. These kernels can be exercised independently. They do not constitute a calibrated climate-to-cost simulation.

3 Independent mathematical reconstruction

3.1 Emissions and climate kernels

For annual emissions q_t in $\mathrm{Gt\,CO_2\,yr^{-1}}$, the target's printed equation is

$$q_t = \begin{cases} q_{t-1} + \gamma_g(\tau - t), & t \le \tau, \\ q_{t-1} - 2q_\tau \gamma_d e^{\gamma_d(t-\tau)}/[1 + e^{\gamma_d(t-\tau)}]^2, & t > \tau. \end{cases} \tag{1}$$

The implementation evaluates the second branch with negative exponential arguments to avoid overflow while preserving the algebra. The growth coefficient is interpreted in the annual recurrence's time convention; a dimensional continuous-time growth coefficient would have different units. An integer peak year is required explicitly. The emissions prior allows continuous peak years, but the treatment of q_τ for a noninteger peak was not recovered.

The 2024 manuscript gives independent truncated-normal priors for growth, peak year, and decline: means $(0.004, 2070, 0.07)$, standard deviations $(0.0075, 25, 0.05)$, and bounds $[0.001, \infty)$, $[2030, 2200]$, and $[0.001, 0.2]$, respectively [2]. These are not the posterior distributions used in the target ensemble. The subsequent inversion uses expert cumulative-emissions information whose

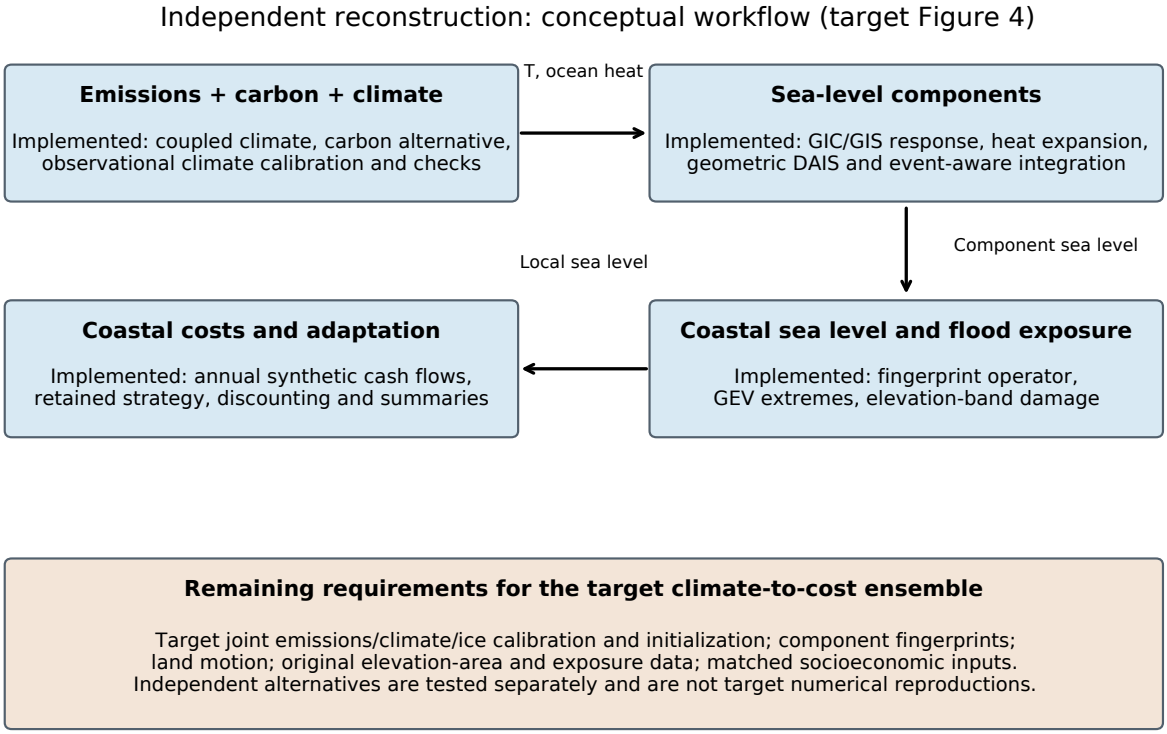


Figure 1: Original conceptual schematic corresponding to target Figure 4. Blue boxes identify implemented component calculations and their physical interfaces. The lower box lists unresolved inputs and model closure. The arrows describe required exchanges, not a claim that a complete coupled ensemble has been executed. This diagram is not counted as a reproduced numerical result.

complete probability-density prescription was not recovered. Sampling these priors would therefore not reproduce the target emissions distribution.

Recoverable carbon kernels are net primary production $N = N_0[1 + \beta \ln(C/C_0)]$, with $N_0 = 60\,\mathrm{Gt\,C\,yr^{-1}}$ and $C_0 = 280\,\mathrm{ppm}$, and heterotrophic respiration $(a_1 D + a_2 S) Q_{10}^{\Delta T/10}$ [4]. Here D and S are detrital and soil carbon, and a_1, a_2 are turnover rates. The continuation recovers terrestrial allocation and carbonate equilibrium, allowing the closed, explicitly alternative calculation in Section7. The implemented CO_2 forcing is $3.7 \log_2(C/C_0)\,\mathrm{W\,m^{-2}}$ [3].

The ocean verification operator solves $\partial_t T = \kappa \partial_{zz} T$ on a one-dimensional column with prescribed surface temperature and zero bottom heat flux. Cell-centred finite volumes use half-cell distance at the upper boundary. Backward Euler time integration ensures that the column heat change equals the top flux evaluated at the new time, to roundoff. Multiplication by volumetric heat capacity converts the tested temperature integral to heat per unit area. This independently chosen discretization is not asserted to match the original SNEASY solver.

3.2 Sea-level components and spatial transformation

The glacier contribution S satisfies the documented reservoir relation

$$\frac{dS}{dt} = \beta(T - T_{\mathrm{eq}})(1 - S/V_0)^n, \qquad T_{\mathrm{eq}} = -0.15\,^{\circ}\mathrm{C}. \tag{2}$$

Temperature is referenced to 1850–1870, V_0 is initial glacier volume expressed as metres of sea-level equivalent, and β has units $\mathrm{m\,K^{-1}\,yr^{-1}}$ [5]. The numerical domain requires $S \le V_0$ for fractional n. An integration step that leaves this domain raises an error; no undocumented volume clipping is introduced.

Greenland volume V, also in metres of sea-level equivalent, follows

$$V_{\mathrm{eq}} = cT + b, \qquad \frac{dV}{dt} = (V_{\mathrm{eq}} - V)(\alpha T + \beta). \tag{3}$$

The temperature reference is 1961–1990, and the volume loss relative to its initial state yields sea-level rise. The rate $\alpha T + \beta$ must be nonnegative. Constant temperature gives an exponential relaxation and therefore an independent verification solution.

The target lineage uses a heat-driven thermal-expansion response, $\Delta S_{\mathrm{TE}} = k_H \Delta H$, not the earlier temperature-relaxation emulator [9, 2]. The coefficient k_H in $\mathrm{m\,J^{-1}}$ remains an explicit input. If the density expansion parameter α_ρ has units $\mathrm{kg\,m^{-3}\,K^{-1}}$, dimensional mass and heat accounting gives $k_H = \alpha_\rho/(AC_p\rho^2)$. This is a dimensional derivation, not recovery of a calibrated numerical coefficient. Supplying a conventional volumetric coefficient in $\mathrm{K^{-1}}$ in that expression would be a unit error.

The documented Antarctic ocean-temperature transformation is

$$T_o = T_f + \frac{x}{1 + e^{-x/a}}, \qquad x = aT + b - T_f, \qquad T_f = -1.8\,^{\circ}\mathrm{C}. \tag{4}$$

An additional threshold-triggered loss rate can be evaluated when its threshold and rate are supplied. The continuation implements DAIS accumulation, marine discharge and differentiated geometry from Shaffer[7], with sign and unit corrections derived explicitly. Present-day equilibrium and warming tests verify these mechanisms. The target paleo calibration, initial state and full coupling remain distinct recovery tasks. Land-water increments remain externally specified.

Local relative sea level is represented by $L_i(t) = \sum_c F_{ic} S_c(t) + v_i(t - t_0)$, where F contains component fingerprints and positive v_i increases relative sea level. This sign convention includes

subsidence as positive. The matrix operation is verified on known area-weighted factors; actual fingerprint grids and coast matching are absent. Reference-period routines require complete annual coverage and refuse silent rebasing of an incomplete series.

3.3 Coastal damage and costs

The independently implemented flooding calculation uses a generalized extreme value (GEV) distribution for annual extreme water level and piecewise uniform area density within supplied elevation bands [8]. For water height w and exposed elevation e, property damage is proportional to $\phi(w-e) = (w-e)/(h_0+w-e)$ with $h_0 = 1\,\mathrm{m}$. The integral over an exposed band $[a, b]$, where $b \le w$, is

$$\int_a^b \phi(w-e)\,de = (b-a) - h_0 \ln\left(\frac{h_0+w-a}{h_0+w-b}\right). \tag{5}$$

Multiplication by area density and capital density yields event loss. Mortality uses flooded population, a supplied mortality fraction and value of statistical life. Both contributions are multiplied by $1-r$, where r denotes economic resilience. Integration of this loss against the GEV density produces expected annual loss. The surge integration bound is explicit, and the truncated density is not renormalized. Retreat changes the lowest retained elevation; protection changes the overtopping threshold.

Relocation combines income-related moving cost, mobile-capital transfer cost, and demolition of immobile capital. The documented benchmark mobile share is one quarter; immobile capital is three quarters. Reactive relocation costs five times the proactive counterpart in the target model. Separate kernels calculate inundation with explicit depreciation, wetland loss with a critical rise rate, and quadratic-height seawall construction. Annual maintenance is returned as a separate cash flow. The numerical coefficient in the construction formula must have units consistent with its quadratic height dependence; the quoted historical reference cost is not silently substituted where its unit convention is ambiguous.

For annual payments C_y, the explicit discount operator is

$$\mathrm{PV}_{y_0} = \sum_y \frac{C_y}{(1+r)^{y-y_0}}. \tag{6}$$

A 2010 base year and 4% rate are used in verification. This base-year convention does not resolve the target caption's wording about discounting through/to 2150. The optimizer chooses the least-cost member of a supplied finite candidate menu. It does not construct the original adaptation trajectories, because their physical and socioeconomic inputs are missing.

4 Verification results

All 80 original Julia assertions pass across seven groups: emissions/carbon/forcing; ocean conservation and convergence; sea-level limits; spatial accounting and references; extreme levels and damage integration; costs and optimization; and ensemble/regional arithmetic (the sea-level group includes ODE accuracy). These original tests are numerical verification tests. Additional mechanism tests and observational checks are described in Section7. The tests require no third-party Julia packages. Configuration values are explicitly marked as manufactured in `config/validation.toml`. Table1 is generated directly from the diagnostic files and the independent SciPy checks.

Verification quantity	Calculated value
Greenland finest absolute error (m SLE)	5.398e-10
Ocean finest maximum temperature error (K)	6.759e-06
Ocean maximum heat-balance residual (K m)	1.021e-16
Expected flood loss, SciPy reference (USD/year)	442422.479342
Expected flood loss, Julia relative difference	1.271e-11
Depth integral maximum difference (USD)	9.313e-10
Annual emissions residual at gamma=0.07 (% of peak)	1.750000
Minimum sampled Antarctic ocean transform (degree C)	-2.078464

Table 1: Computed verification metrics. These quantities describe manufactured test cases, not target global projections. The heat-balance residual is reported before multiplication by volumetric heat capacity.

4.1 Printed emissions recurrence and its limit

Figure2 compares the literal recurrence with the continuous logistic decline $q(t)/q_\tau = 2/[1 + e^{\gamma_d t}]$. The decrement in Equation1 is the right-endpoint sample of the continuous decline rate. Summing endpoint samples does not exactly integrate that rate. For small step Δt, the residual fraction is $\gamma_d \Delta t/4$ to leading order. At an annual step and decline rate $0.07\,\mathrm{yr}^{-1}$, the calculated residual is approximately 1.75% of peak emissions. The tested rates 0.01 and $0.2\,\mathrm{yr}^{-1}$ give approximately 0.25% and 5%, respectively. Refining the quadrature step reduces the residual. These are normalized mathematical cases extending 2000 years after an arbitrary peak, not calendar-year emissions projections.

The discrepancy separates two interpretations of the published specification: literal annual accumulation and a continuous logistic trajectory that vanishes asymptotically. No choice was imposed on a global simulation. The effect on temperature and costs cannot be calculated without the missing coupled model.

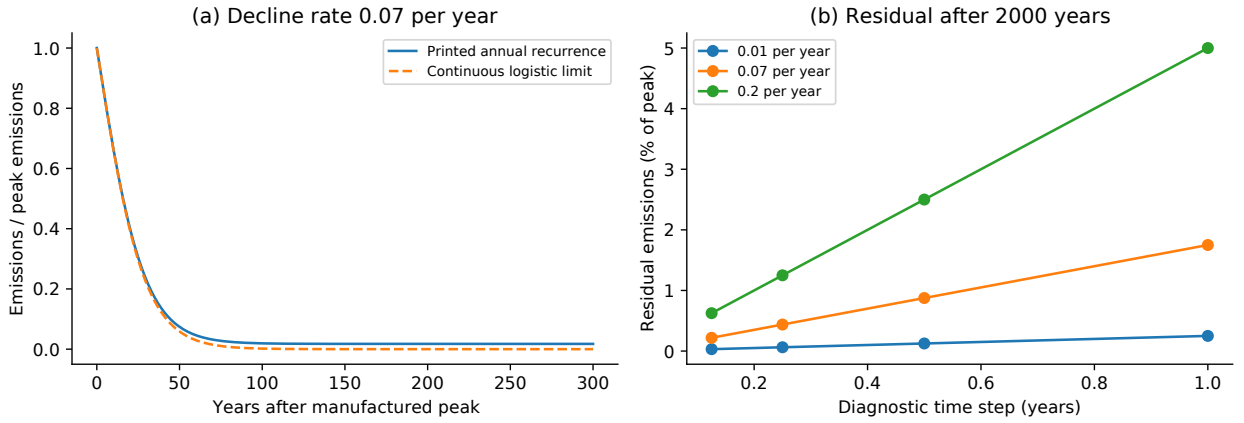


Figure 2: Independent diagnostic of the printed emissions equation. (a) Annual endpoint accumulation for normalized peak emissions and decline $0.07\,\mathrm{yr}^{-1}$, compared with the continuous logistic expression. (b) Remaining emissions after 2000 years for three decline rates and four diagnostic step sizes. The residual decreases with step size; this does not establish which interpretation the original software used.

4.2 ODE and ocean-diffusion convergence

Figure3 tests the Greenland relaxation against its exact constant-temperature solution. The manufactured case has $T = 2\,\mathrm{K}$, $c = -0.2\,\mathrm{m\,K^{-1}}$, $b = 1\,\mathrm{m}$, $\alpha = 0.005\,\mathrm{K^{-1}\,yr^{-1}}$, $\beta = 0.01\,\mathrm{yr^{-1}}$ and initial volume 1.2 m. The equilibrium is 0.6 m and the integration lasts 100 years. Fourth-order convergence is obtained when the RK4 step is halved.

The ocean case uses a 1 m column, diffusivity $0.1\,\mathrm{m^2\,s^{-1}}$ and initial temperature $\sin[\pi z/(2L)]$ K. The upper temperature is zero and the bottom flux vanishes. The exact amplitude decays as $\exp[-\kappa(\pi/2L)^2 t]$. Time steps decrease with squared grid spacing, so the first-order temporal and second-order spatial errors converge together at second order in spacing. These deliberately artificial dimensions isolate the numerical operator from Earth-system calibration.

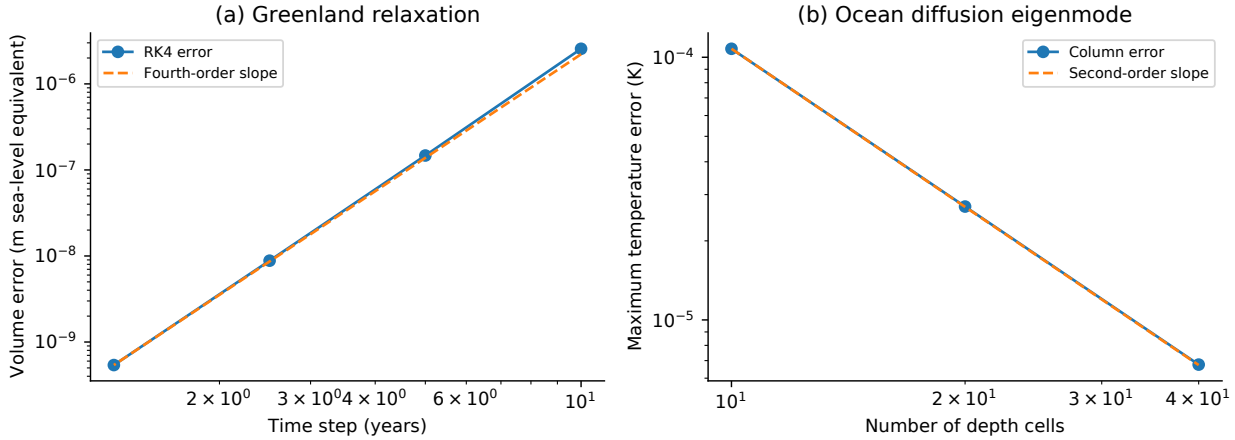


Figure 3: Independent convergence tests. (a) Greenland volume error at 100 years against the exact exponential relaxation; the dashed line has fourth-order slope. (b) Maximum ocean temperature error at 0.2 s for 10, 20 and 40 cells; the dashed line has second-order slope. Time step is 0.2 divided by the square of the cell count, in seconds.

4.3 Flood-loss integration and decision sensitivity

Figure4 uses a manufactured $5\,\mathrm{km^2}$ segment distributed uniformly over 0–5 m elevation. Capital density is one million USD $\mathrm{km^{-2}}$, population density $100\,\mathrm{km^{-2}}$, mortality fraction 0.01, value of life 100,000 USD and resilience 0.2. These values are verification fixtures, not recommended valuations. The GEV parameters are location 1 m, scale 0.3 m and shape 0.1. The outer integral extends from -2 to 20 m. SciPy uses its opposite-sign shape convention and independent adaptive integration for both elevation and surge. This verifies the Julia closed-form depth integral and composite-Simpson expectation.

Table2 quantifies two accounting ambiguities with manufactured costs. A two-period menu has candidate costs $(1, 9)$ and $(6, 2)$ at years 2010 and 2060. Keeping one candidate differs from allowing a switch. This menu has no inherited infrastructure state; it is a test of the decision rule only. Likewise, ten annual unit payments during 2010–2019 differ from paying ten units at either end. A coarse timestep must retain the annual-flow interpretation if it is to reproduce the same present value.

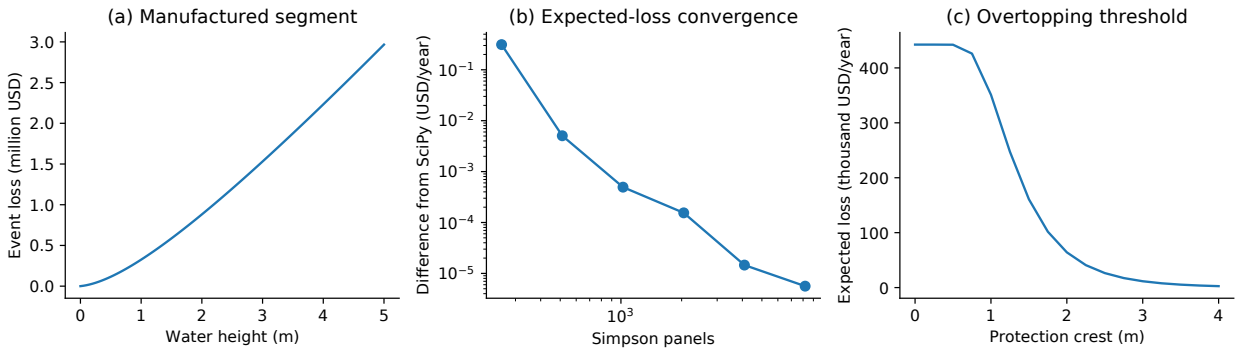


Figure 4: Manufactured coastal damage verification. (a) Event damage increases with water height. (b) Expected annual loss approaches an independent nested SciPy integral as the number of Simpson panels increases. (c) Expected overtopping loss decreases with protection crest. Panel(c) omits construction cost and therefore is not an optimal seawall-height calculation.

Manufactured accounting choice	Present value
fixed strategy pv	2.266414
revisable strategy pv	1.281425
ten annual unit payments pv	8.435332
ten units paid at start pv	10.000000
ten units paid at end pv	7.025867

Table 2: Sensitivity to strategy persistence and payment timing at 4% annual discounting. Units are arbitrary currency; these are not coastal estimates.

4.4 Antarctic transform and its asserted floor

For $a > 0$ and $x < 0$, the added term in Equation4 is negative. The literal transformation therefore falls below T_f, despite its description as bounded below by freezing [5]. For the test values $a = 1, b = 0$, the minimum is approximately $-2.07846\,^{\circ}\mathrm{C}$. In dimensionless form, $z = x/a$, the extremum satisfies $1 + e^{-z} + ze^{-z} = 0$, giving $z \simeq -1.27846$ and a minimum increment $-0.27846a$. Figure5 shows this algebraic behavior. No freezing clamp was added. Whether the calibrated historical trajectory visits this regime remains unknown.

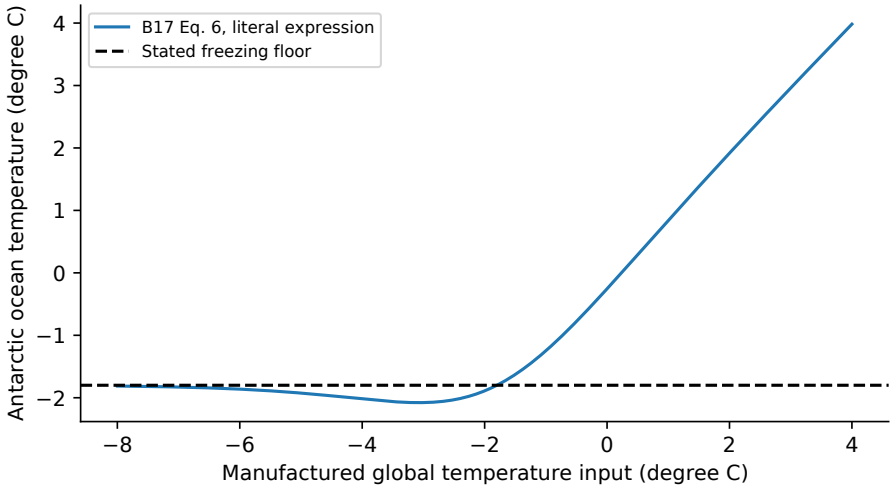


Figure 5: Literal BRICK Antarctic ocean-temperature transformation with manufactured coefficients $a = 1, b = 0$ and stated freezing temperature $-1.8\,^{\circ}\mathrm{C}$. The curve falls below the dashed floor for negative x. This is a specification diagnostic, not an Antarctic projection.

5 Quantitative comparison boundary

The target reports median costs of 1.2, 1.5, 2.1 and 2.6 trillion 2010 USD across the four warming bins, with fifth percentiles 1.0, 1.2, 1.6 and 2.2 trillion USD [1]. The prose supplies 95th percentiles 1.3 and 2.3 trillion USD for the first two bins; unreported numerical endpoints were not digitized or invented. No independent global costs exist here, so neither relative errors nor agreement intervals can be calculated for those targets.

An arithmetic audit of the printed Table 1 is possible. Its four counts sum to 5,000. Among 250 simulations above 2.7 trillion USD, the text gives 125 in the highest bin and 109 in the third, with none in the lowest. The remaining 16 belong to the second bin. Dividing these counts by bin size yields the rounded probabilities in the paper (Table3). This verifies internal consistency of printed summaries; it does not reproduce the modeled probability distribution or establish an independently calculated tail threshold.

6 Complete figure-by-figure record

Table4 covers all main and supplementary figures and the main table. Unlettered joint and marginal views are listed separately. Every entry's dependencies, source provenance, assumptions, commands, comparison status and failure explanation are also recorded in `docs/FIGURE_STATUS.csv`. For

Bin	Printed count	Tail count	Calculated%	Printed%
1	202	0	0.0000	0
2	2264	16	0.7067	1
3	2145	109	5.0816	5
4	389	125	32.1337	32

Table 3: Arithmetic audit using published counts only. The second-bin tail count 16 is derived by subtraction. These data never enter the model kernels.

unavailable outputs, the generation command is explicitly unavailable; there is no plotting command that can turn missing model predictions into an independent reproduction. Supplementary Figure 10 is one histogram containing all-member and Antarctic-trigger-conditioned distributions of proactive migrants.

Table 4: Target reproduction status. "Not reproduced" means no independently generated target numerical result.

Figure/view	Quantity	Status
1: global interval plot	Conditional median and 5th-95th cost quantiles	Not reproduced
2: joint scatter	Sea level 2100 versus discounted costs	Not reproduced
2: top marginal	Sea level distribution by warming bin	Not reproduced
2: right marginal	Cost distribution by warming bin	Not reproduced
3: regional bars	Regional cost/GDP minus global cost/GDP	Not reproduced
4: workflow	Model components and exchanges	Conceptual schematic
S1: interval plot	5000 versus 4000 subsample comparison	Not reproduced
S2: interval plot	202 per bin versus full sample	Not reproduced
S3: trajectory plot	GMST median and 90% intervals plus AR6 boxes	Not reproduced
S4: trajectory plot	GMSL median and 66% intervals plus AR6 boxes	Not reproduced
S5: map	Median local sea level 2100	Not reproduced
S6: interval plot	GTSR versus DINAS cost intervals	Not reproduced
S7: joint scatter	Sea level versus costs above 2.7 trillion	Not reproduced
S7: top marginal	Upper-tail sea level distribution	Not reproduced
S7: right marginal	Upper-tail cost distribution	Not reproduced
S8: regional bars	SSP2 regional cost/GDP deviations	Not reproduced
S9: regression plot	Regional SSP income contrast versus cost sensitivity	Not reproduced
S10: histograms	Migration conditional on AIS trigger	Not reproduced
Table1: table	Warming-bin counts and tail probabilities	Not reproduced

Main Figure 1 requires full conditional cost distributions. Figure 2 requires joint sea-level/cost realizations, and Figure 3 additionally requires regional economic aggregation. Supplementary Figures 1 and 2 require controlled subsampling of a legitimate ensemble, not merely a sample-size calculation. Figures 3 and 4 require calibrated time series plus AR6 comparisons. Figure 5 requires spatial fingerprints and site matching. Figure 6 requires matched GTSR and DINAS extreme levels. Figure 7 requires the original type of upper-tail joint distribution. Figures 8 and 9 require paired SSP2/SSP4 economics. Figure 10 requires retained segment migration histories and Antarctic-trigger states. The implemented summary and aggregation routines do not supply these missing physical inputs.

7 Continued recovery, implementation and observational experiments

The earlier assessment identified missing mechanisms too broadly. Reading the complete Tanaka thesis and the 2022 scholarly preprint recovered the DOECLIM constants, the coastal decision menu and the socioeconomic priors. Independent providers supplied forcing, temperature, carbon

7.1 Coupled energy balance and numerical heat transport

The land–troposphere and ocean–troposphere boxes exchange heat and lose energy through separate radiative feedbacks [14, 3]. For land fraction $f = 0.29$, their temperatures T_L, T_S satisfy

$$C_L \dot{T}_L = Q - \lambda_L T_L - \frac{k}{f}(T_L - bT_S), \tag{7}$$

$$C_S \dot{T}_S = Q - \lambda_S T_S + \frac{k}{1-f}(T_L - bT_S) - F_O. \tag{8}$$

Here $b = 1.3$, $C_L = 0.52$ and $C_S = 7.8\,\mathrm{W\,yr\,m^{-2}\,K^{-1}}$. The global surface-air observable is $fT_L + (1-f)bT_S$. The equilibrium land/SST ratio is 1.43 and $k = 1.59 - 0.31\lambda_L$. Together with an equilibrium warming of ECS under $3.7\,\mathrm{W\,m^{-2}}$, these relations determine both feedbacks and the exchange coefficient. The explicit formulas are recorded in `docs/RECOVERED_SPECIFICATIONS.md`.

An 80-cell, 4000 m interior ocean exchanges heat continuously with the mixed layer and has an insulated bottom. Finite-volume fluxes cancel under a global heat-capacity weighted sum; this is tested independently of equilibrium fitting. Crank–Nicolson integration uses four steps per year. Refining the vertical grid from 80 to 160 cells changes the year-70 warming under 1% annual CO_2 growth by 0.000787 K; the next doubling changes it by 0.000197 K. This approximately fourfold error reduction agrees with second-order spatial convergence.

The 60 m effective mixed-layer capacity contains about 2.5 m of atmospheric-equivalent capacity. Comparisons with NOAA integrate only 57.5 m of water and the appropriate fractional interior cells down to the stated observational depth. Whole-column heat is not compared to a 700 m record. HadCRUT4 blends land air temperature with SST, whereas the legacy DOECLIM observable includes enhanced marine-air warming. Treating that blended record as the global-air observable follows the simplified legacy calibration and is an additional representation approximation, not an exact observation operator.

7.2 A closed, explicitly alternative carbon cycle

The recovered terrestrial stocks are 100, 500, 120 and 1500 Gt C in leaves, wood, detritus and soil [15]. Baseline NPP of $60\,\mathrm{Gt\,C\,yr^{-1}}$ allocates 35 to leaves and 25 to wood. Leaf turnover sends 35 to detritus; wood sends 20 to detritus and 5 to soil. Detritus respires 45 and transfers 10 to soil; soil respires 15. Linear turnover rates are these fluxes divided by their source stocks. Fertilization modifies NPP and the documented Q_{10} factor modifies the two respiration fluxes [4]. All internal transfers cancel in the global carbon balance.

Ocean carbon uses an instantaneously equilibrated atmosphere–mixed-layer reservoir and three deeper reservoirs. Constant alkalinity, total boron and carbonate equilibrium determine atmospheric CO_2 and mixed-layer carbon from their combined inventory [16]. The nonlinear scalar root is solved by bisection in log hydrogen concentration. At 280 ppm the reconstructed DIC is $0.002080909\,\mathrm{mol\,L^{-1}}$, consistent with the printed H99 value to its reported precision. The mixed-layer depth and deep exchange parameters are from the later Tanaka description. Mixing these documented versions, and holding carbonate chemistry temperature fixed, defines an alternative closure; it does not establish exact recovery of the target SNEASY version.

Second-order splitting couples carbon RK4 steps to climate CN steps. Twenty-one checks establish baseline equilibrium, chemical partition, stock positivity, second-order coupled convergence and total-carbon conservation. Across 18 historical sensitivities, the largest carbon-budget residual is below 10^{-11} Gt C. Conservation alone does not validate the response: the uncalibrated default predicts 461.38 ppm in 2022, compared with NOAA's 417.08 ppm. The GCB historical table has no

pre-1850 land-use emissions; the input retains blanks, while separate calculations prescribe zero or the 1850 rate before 1850. These are alternatives, not recovered observations.

7.3 Temperature-only calibration and failed holdout checks

The first observational experiment estimates ECS, heat diffusivity, aerosol multiplier, temperature offset, AR1 innovation scale and AR1 correlation. HadCRUT4.6.0.0 years 1850–2005 form the training data; years 2006–2021 are held out. Known observational standard deviations are approximated by the total 95% interval widths divided by 2(1.96). This approximation omits the record's full temporally correlated measurement covariance. A scalar Kalman filter integrates the latent AR1 residual exactly under these assumptions; tests match the dense Gaussian covariance likelihood to 10^{-12}.

Uniform bounds follow the B17 climate table, except that ECS is restricted to 0.7–10 K: the recovered exchange relation becomes negative below about 0.68 K. CEDA effective forcing supplies historical and future inputs, with both aerosol components multiplied by the same parameter. This differs from the older indirect-aerosol convention and from Darnell's uncertain emissions and nearest-SSP forcing construction. Neither the posterior nor the six scenario experiments is claimed as the Wong ensemble.

An initial short run failed convergence checks. The retained calculation uses four dispersed chains, each with 6000 warmup and 30,000 production steps; every fifth state is retained. Proposal adaptation stops before production. Log ECS and log diffusivity proposals include their Jacobian, preserving the physical-coordinate prior. Independently implemented rank/folded split $\widehat{R}$ and bulk/tail ESS follow Vehtari et al. [17]. The largest $\widehat{R}$ is 1.00427 and the smallest ESS is 1894, satisfying the prespecified limits of 1.01 and 400.

Conditional forecasts propagate the final training-period residual state and include both AR1 innovation and observational uncertainty. Despite sampler convergence, holdout temperature RMSE is 0.209 K, with mean bias +0.189 K. Only 10 of 16 held-out annual temperatures lie inside the nominal 90% predictive intervals. After fitting only an additive heat datum offset over 1955–2005, the 700 m heat parameter-median RMSE in 2006–2021 is 16.46×10^{22} J. These failures motivate a separate temperature–heat calibration; they are not removed from the record.

One thousand complete joint parameter rows generate six SSP trajectories; 250- and 500-row subsets test sampling variation. The physical temperature intervals exclude future residual noise. SSP2-4.5 mean warming over 2081–2100 is 2.924 K, with a parameter 90% interval of 1.976–4.188 K. The corresponding printed AR6 assessment is 2.7 [2.1, 3.5] K. This comparison uses the same averaging/reference periods but compares different inference systems; it is not a numerical reproduction of Supplementary Figure S3.

7.4 Antarctic events and observational thermal expansion

DAIS geometry, accumulation and marine discharge are now implemented. The discharge water depth follows sea level minus bed elevation, correcting the printed sign from geometry; coefficient units follow dimensional consistency. The equilibrium volume of $2.478479 \times 10^{16}\,\mathrm{m}^3$ differs from Shaffer's rounded $2.478 \times 10^{16}\,\mathrm{m}^3$ by 0.0193%. This is a reproduced methodological benchmark, not a reproduced Wong figure.

The printed runoff gate activates at $h_0 + cT_A = 0$. Over marine bed this can produce a discontinuous accumulation rate. Splitting time steps at that event and evaluating endpoint stages on the appropriate side removes the earlier irregular refinement. In the 6 K, century-long warming ramp, the event occurs at year 41.92982456. Two-year and one-year steps differ in year-200 radius

by 1.57×10^{-7} m; finer steps reach floating-point limits. The dashed curves in Figure8 instead activate runoff when the runoff line exceeds local edge height, and remain explicit alternative physics. These tests express sea level by mass conservation using ice/water densities and ocean area, whereas B17 scales its initial Antarctic volume to 57 m. The two observable conventions are not silently interchanged.

An independent coefficient check uses NOAA heat and thermosteric sea level from the same depth range. OLS over 1955–2005 at 0–700 m gives 1.392×10^{-25} m J^{-1}; fitting 2005–2015 at 0–2000 m gives 1.107×10^{-25} m J^{-1}. Held-out thermosteric RMSEs through 2021 are 0.414 and 0.757 mm, respectively. First-difference slopes provide a check against merely fitting a common trend. Because heat and sea level share hydrographic observations, these are descriptive constraints rather than independent-error coefficient posteriors. They are not substituted silently for the unavailable V19 prior or a full-depth expansion coefficient.

7.5 Coastal cash flows and input quality

The annual coastal experiment retains the documented ten-choice menu and first-period decision rule. Sixty checks establish nonnegative cash-flow components, first-period policy selection, no repeated relocation at fixed sea level and telescoping incremental dike construction costs. Upgrades at 2010, 2051 and 2101, annual payment timing and a paired no-SLR counterfactual are explicit alternatives to the target's decadal accounting and initial-step subtraction. Forty-eight synthetic scenarios compare income/exposure, sea-level rise, retreat land accounting and linear versus quadratic maintenance. All exposure values are labeled fixtures; no global cost estimate is inferred. Doubling flood quadrature from 512 to 1024 panels changes the verification-case NPV by about 0.00024 USD and leaves its policy unchanged.

The independent GTSR shapefile has 12,148 coastal feature rows but only 7868 distinct station labels. The initial interpretation of these labels as unique DIVA identifiers was corrected by this audit. A second public release provides 16,611 station-wise Gumbel fits. Geographic mapping leaves 133 nearest matches with invalid parameters; these remain missing, not filled from distant stations. The two input products also disagree in their 100-year levels. These findings prevent a silent merge or assumption of exact input equivalence. The full elevation-area, exposure, country mapping and DINAS tide records still require recovery before geographically aggregated coastal costs can be generated.

Every alternative here has its own input, script, output and figure record. No alternative curve is entered into the target inventory as a recovered published numerical result. Ongoing calibration and data-recovery tasks remain active rather than being reclassified as irreducible merely because the first implementation lacked them.

8 Unresolved inputs and implications

The joint parameter distribution is a central missing scientific object. The cited methodology combines paleo Antarctic constraints, modern climate and component-sea-level observations, uncertain emissions, and autoregressive residuals. Complete target likelihood definitions, data vintages, measurement errors, initial states, and residual treatment were not recovered. Reusing published posterior samples is prohibited here. Independent calibration is now being performed with versioned exogenous observations and explicitly alternative likelihood choices. Independent draws from older marginal quantiles would discard physically relevant dependence.

The historical DIVA exposure bundle is another independent obstacle. The CIAM supplement reports that its v1.5.5 database was no longer supported and that v1.8 was not public at that time

[8]. The EEA provider directory currently reports unavailable software [11]. This does not prove that every DIVA dataset is inaccessible; it establishes that a complete permissible target bundle was not recovered in this investigation. No modern coastline segmentation was substituted for the 12,148 target segments.

Independent GTSR data and six SSP forcing inputs have been acquired [12, 13]. The GTSR release supplies coastal features and a later release supplies Gumbel parameters, but invalid records, repeated station labels and return-level discrepancies require explicit treatment. The 2022 academic supplement was recovered in the scholarly preprint, resolving the six socioeconomic marginal priors and several decision conventions. Country economics, original elevation-area/exposure data, component fingerprints and the precise target data vintages remain unresolved.

The active stages are therefore to extend the independent calibration, reconcile the remaining carbon/ice variants, and recover permissible segment and economic inputs. Small coupled experiments precede any target-sized ensemble. No result in this report supports a quantitative conclusion about the target's regional cost inequality.

9 Assessment of the research-agent process

The agent recovered much of the mathematical interface from distributed scholarly sources, distinguished versions of a mutable preprint, and implemented several independently checkable numerical components. The analytical tests separate solver accuracy from calibration and data availability. Cross-language flood integration supplies an additional check against transcription or implementation errors in the elevation primitive and GEV convention.

Two interpretation errors were avoided through source comparison. First, the older BRICK temperature-based expansion emulator is not the heat-based component described in the target lineage. Second, a later revision behind the same DOI cannot automatically be treated as the manuscript available before publication. Direct calculation then identified two tensions between printed equations and their prose descriptions. These checks do not reveal the original implementation.

The principal limitation is evidence recovery, not numerical throughput. An agent can write functioning code while failing to identify the scientific experiment if it supplies convenient values for missing inputs. This project does not count such code as a successful reproduction. The initial assessment stopped too early: further scholarly reading recovered major mechanisms and independent inputs. Observation-based calibration, carbon mass closure and DAIS integration are now demonstrated, while a global adaptation ensemble remains unreproduced. The failed holdout checks show why solver verification and posterior convergence cannot substitute for validation of the predicted observable.

The research log records source access failures, incidental search exposure, interpretation decisions, and verification outcomes. No subagents were used. The work followed Brad Lipovsky's supplied research instructions and academic style guide. This assessment concerns the recorded execution; it is not a general benchmark of all AI research systems.

10 Reproducibility and current scope

Run `bash scripts/run_all.sh` from the project root to execute the offline verification, generate numerical diagnostics and figures, and compile this report. Julia uses standard libraries only. Python 3.6.8, NumPy 1.19.5, SciPy 1.5.4 and Matplotlib 3.3.4 were available locally; pdfLaTeX comes from TeXLive 2018. The driver verifies executable availability and imports, requires no installation, and

limits numerical libraries to one thread. Calculations run at scheduling priority 19 on a heavily loaded shared server. All artifacts, temporary files and tool caches are directed into the designated project. Source acquisition is a separate optional, hash-checked command. The scientific build does not require network access or downloaded authors' outputs.

The independent calculations verify response and cost mechanisms, quantify written-specification discrepancies, and test alternative observation-conditioned climate scenarios. An original conceptual Figure 4 is complete. Main Figures 1–3, supplementary Figures 1–10 and modeled Table 1 remain unreproduced. A printed-table arithmetic agreement is not promoted to numerical-model agreement. The missing calibration and coastal inputs must be resolved before an assessment of the target scientific conclusions can be claimed. These remaining tasks do not invalidate the completed mechanism tests or authorize treating this progress report as project completion.

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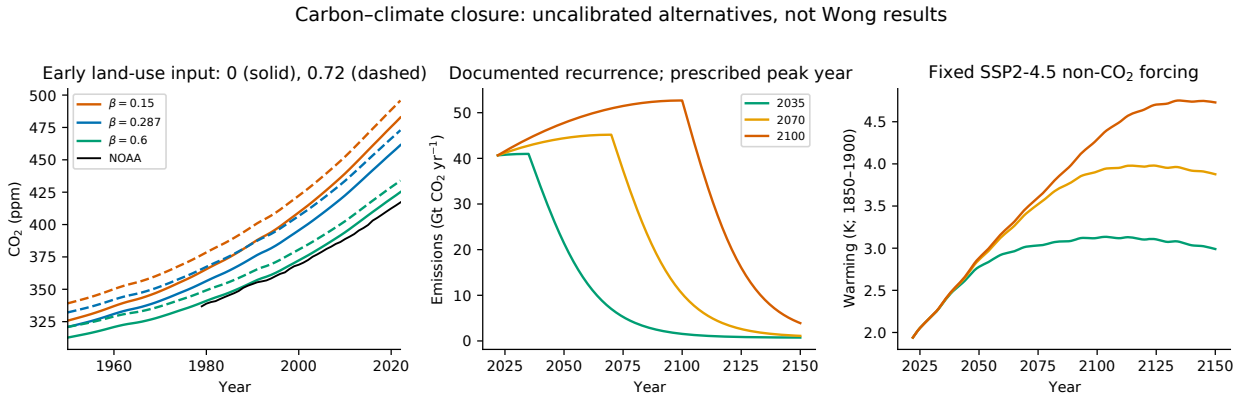


Figure 6: Uncalibrated carbon–climate experiments, not target reproductions. Left: historical CO_2 sensitivities to fertilization and the explicitly assumed early land-use input, with independent NOAA observations. Centre: documented emissions recurrence initialized from GCB2023, with three prescribed peak years. Right: coupled warming under fixed SSP2-4.5 non-CO_2 forcing. The historical mismatch limits interpretation of these future curves.

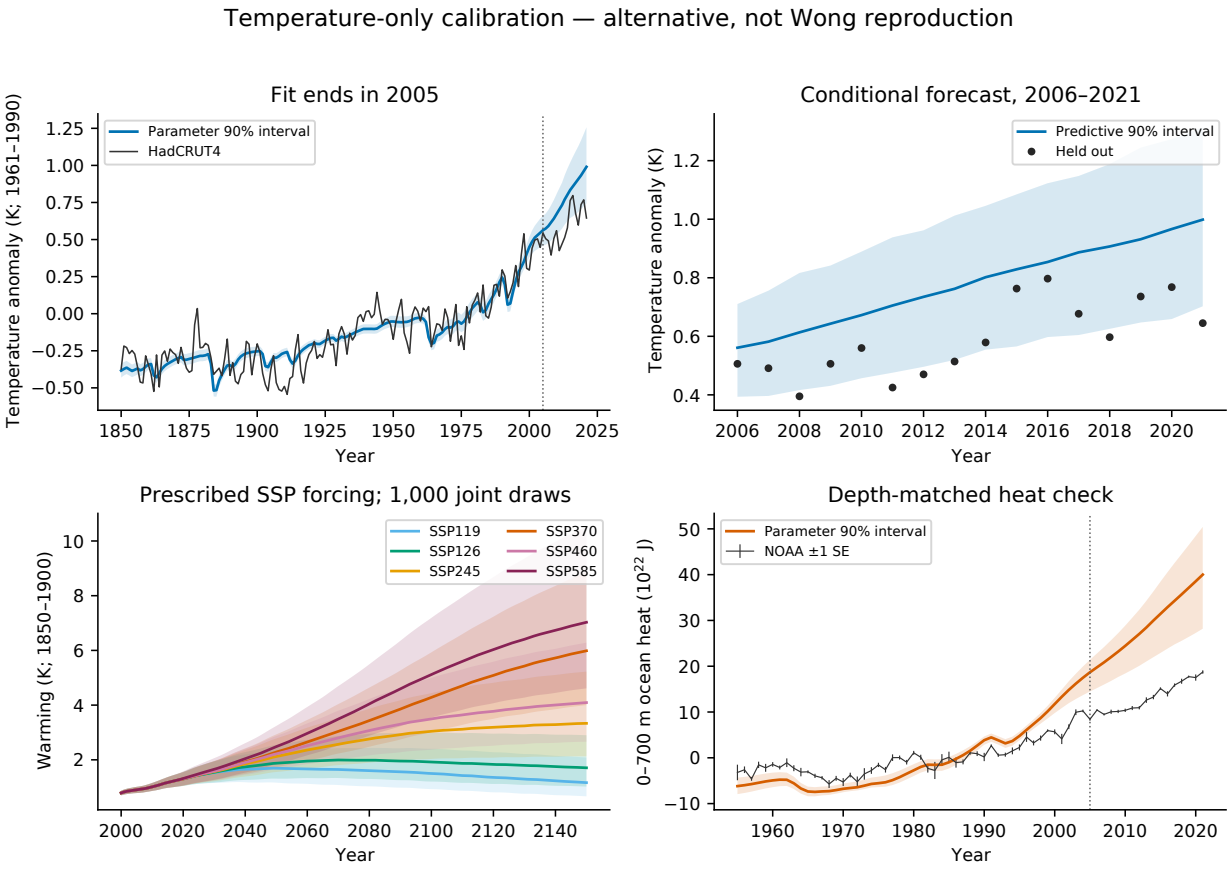


Figure 7: Temperature-only calibration and independent observational checks. The upper-right forecast includes residual and measurement uncertainty. The lower-left scenario bands contain parameter uncertainty only. Ocean heat is compared at the actual observational depth, with a training-period datum offset. The temperature and heat holdout discrepancies prevent treating this converged alternative posterior as validation of the target model.

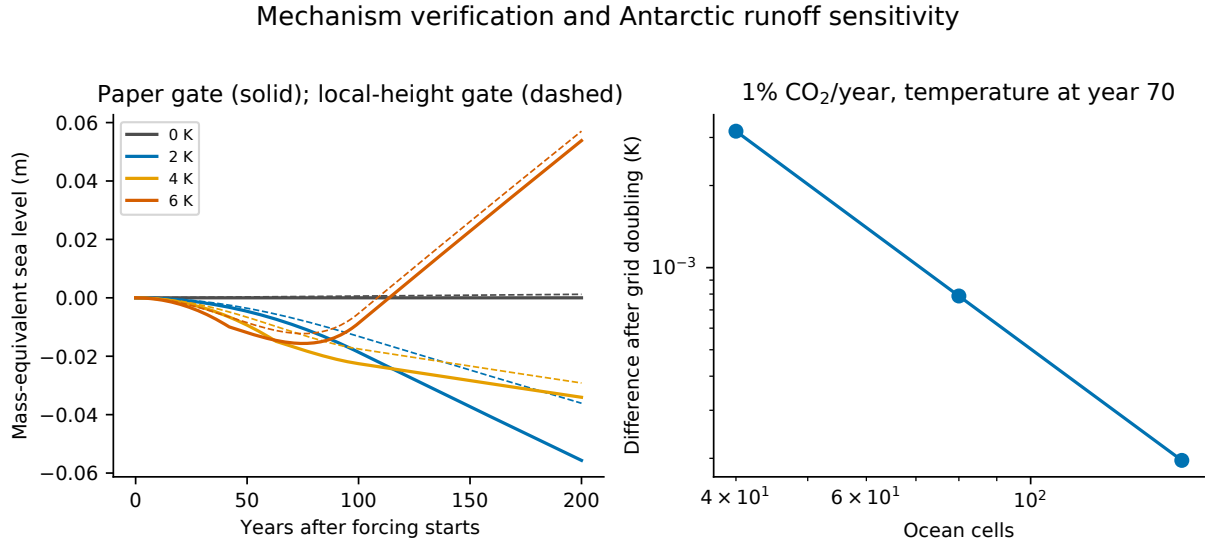


Figure 8: Independent mechanism tests. Left: prescribed Antarctic warming, with fixed ocean temperature and sea level, isolates the runoff convention. Right: climate grid refinement under 1% annual CO_2 growth. Neither panel is a global calibrated projection.

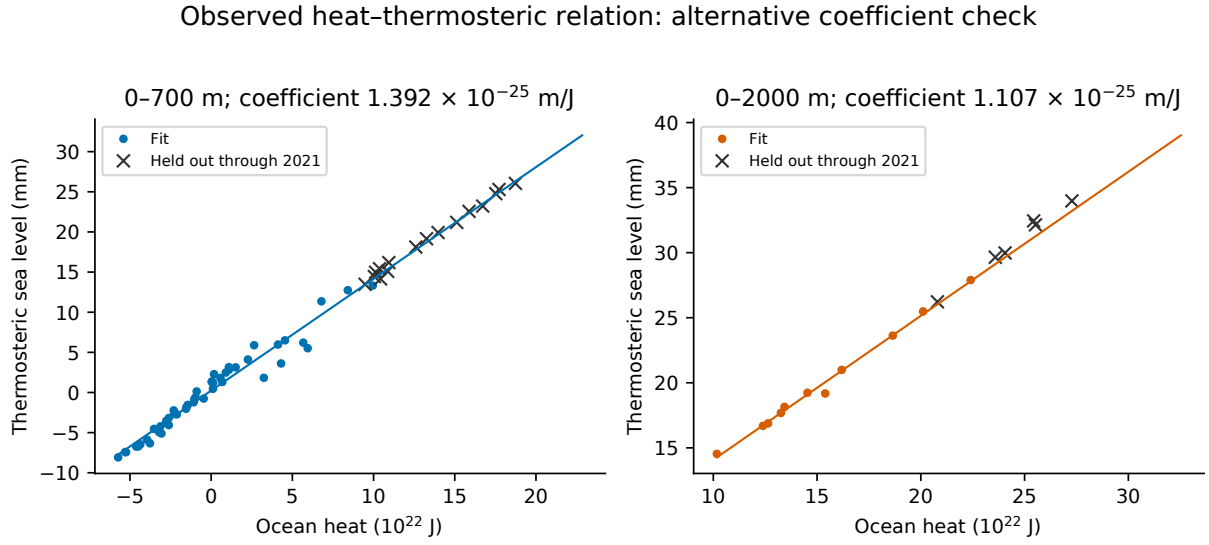


Figure 9: Independent NOAA heat and thermosteric observations with fitted depth-specific linear relations. Later years through 2021 were withheld. This coefficient check is an explicit alternative to the target calibration.

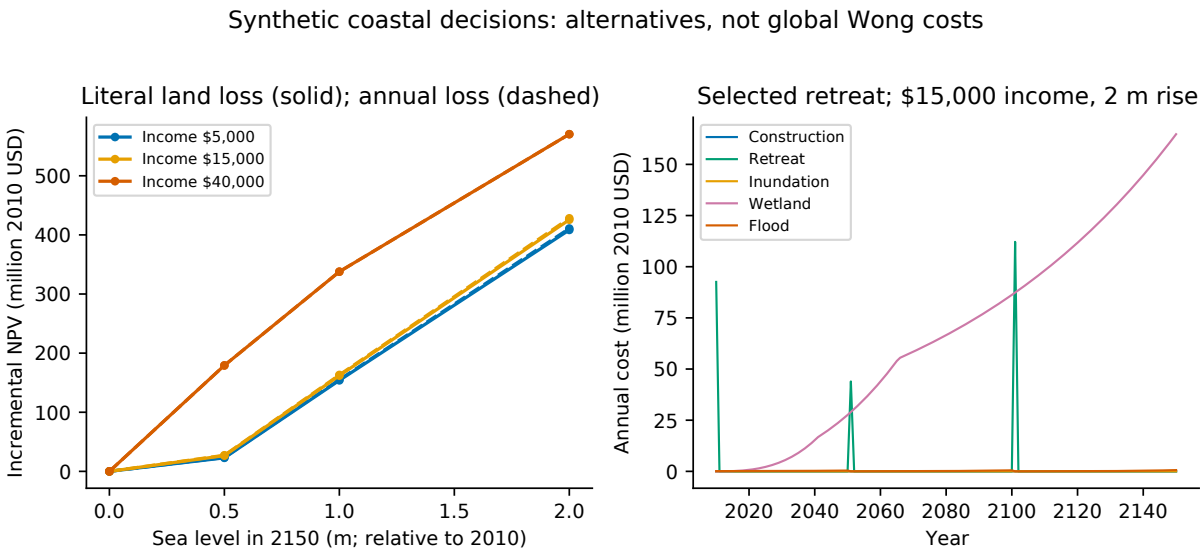


Figure 10: Synthetic coastal alternatives. Left: paired-counterfactual incremental costs under two explicit land-accounting rules. Income and exposure both vary across the three fixtures, so this is not an isolated income effect. Right: annual components for one selected retreat policy, including discrete upgrades. These outputs do not estimate a named coast or a global cost.